

Computer Communication & Networks

Lecture 7

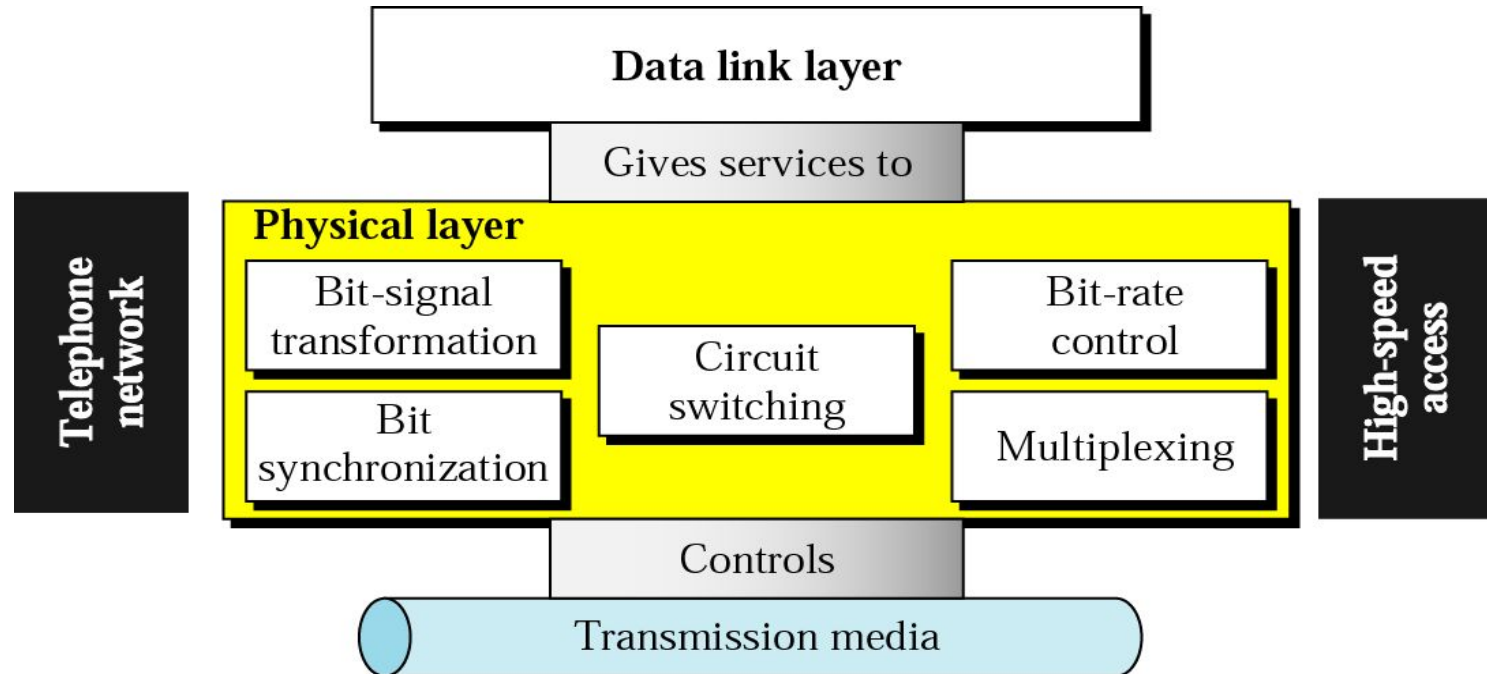
Physical Layer: Analog Transmission

<http://web.uettaxila.edu.pk/CMS/coeCCNbsSp09/index.asp>

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Physical Layer



Physical Layer Topics to Cover

Signals

Digital Transmission

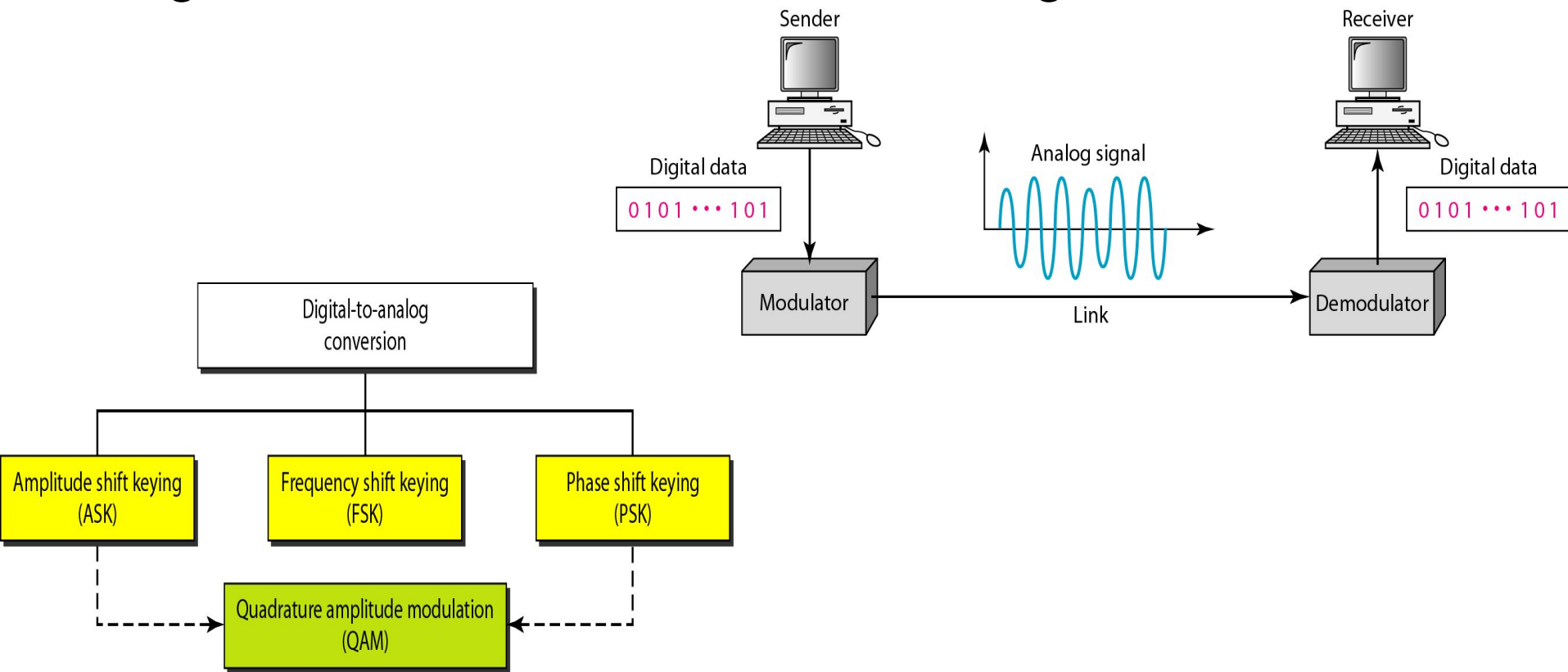
Analog Transmission

Multiplexing

Transmission Media

DIGITAL-TO-ANALOG CONVERSION

- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.



Note

**Bit rate is the number of bits per second.
Baud rate is the number of signal
elements per second.**

**In the analog transmission of digital
data, the baud rate is less than
or equal to the bit rate.**

Example

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

Solution

In this case, $r = 4$, $S = 1000$, and N is unknown. We can find the value of N from

$$S = N \times \frac{1}{r} \quad \text{or} \quad N = S \times r = 1000 \times 4 = 4000 \text{ bps}$$

Example

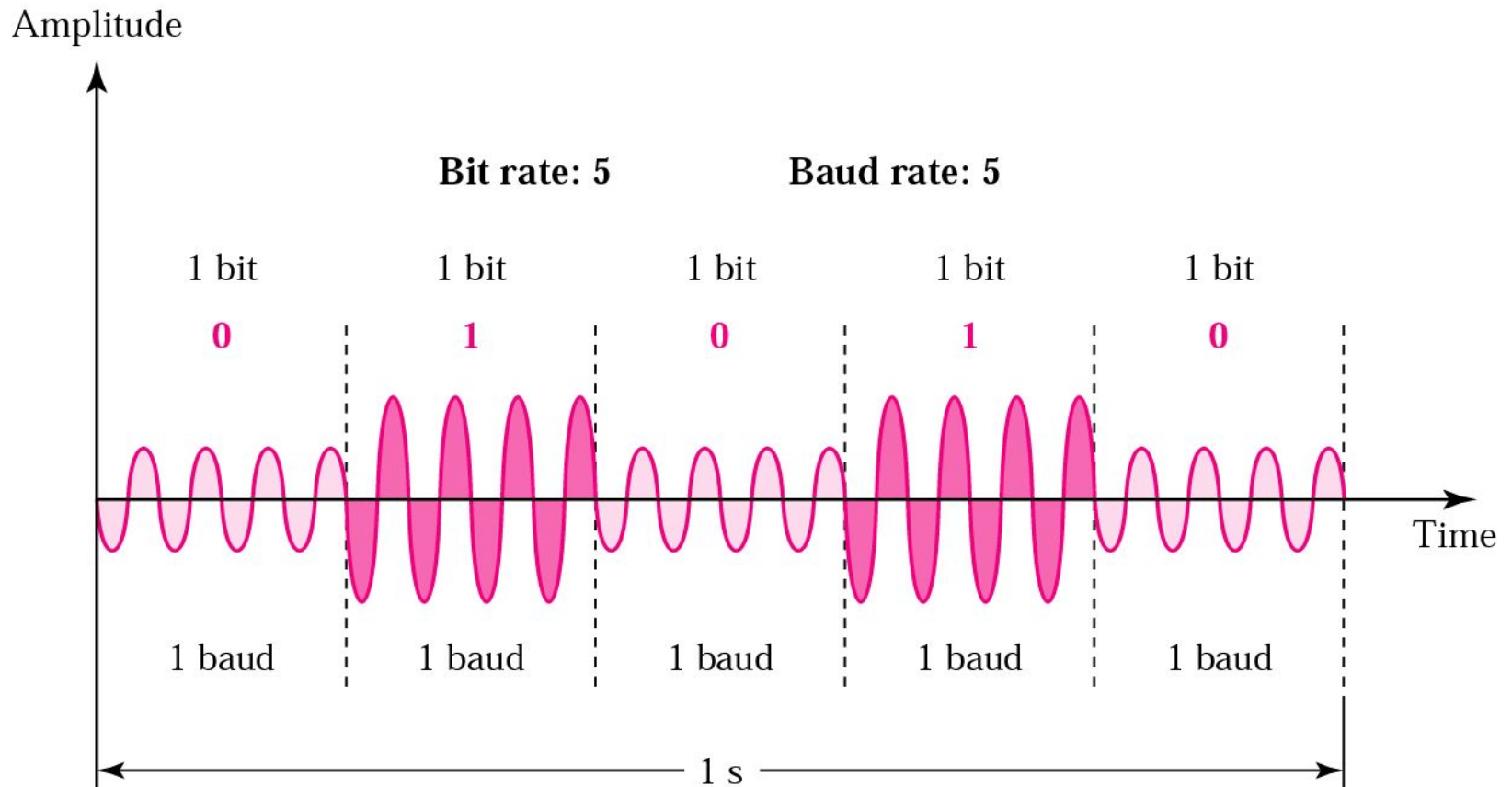
An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

Solution

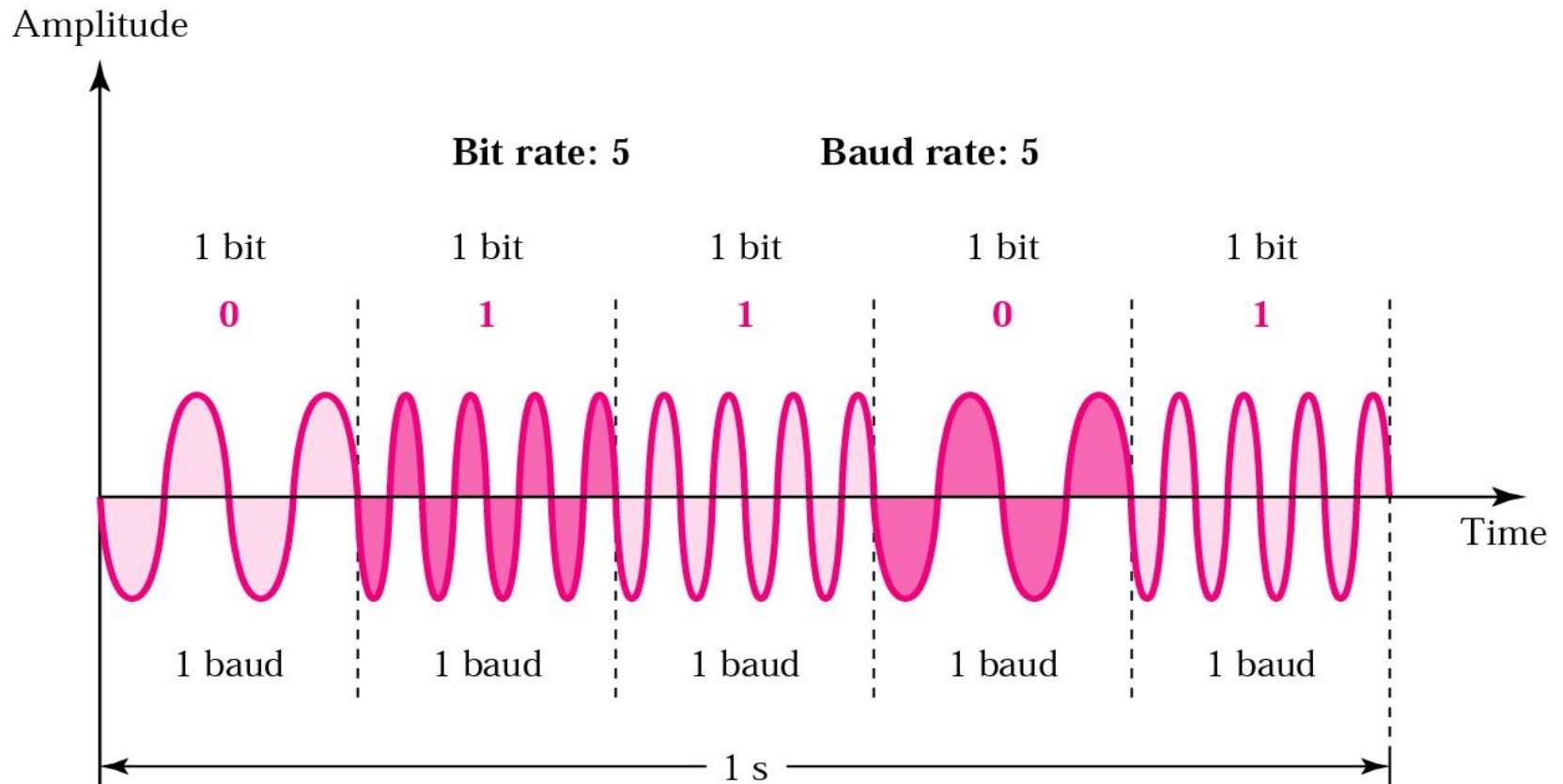
In this example, $S = 1000$, $N = 8000$, and r and L are unknown. We find first the value of r and then the value of L .

$$\begin{aligned} S &= N \times \frac{1}{r} & \longrightarrow & \quad r = \frac{N}{S} = \frac{8000}{1000} = 8 \text{ bits/ baud} \\ r &= \log_2 L & \longrightarrow & \quad L = 2^r = 2^8 = 256 \end{aligned}$$

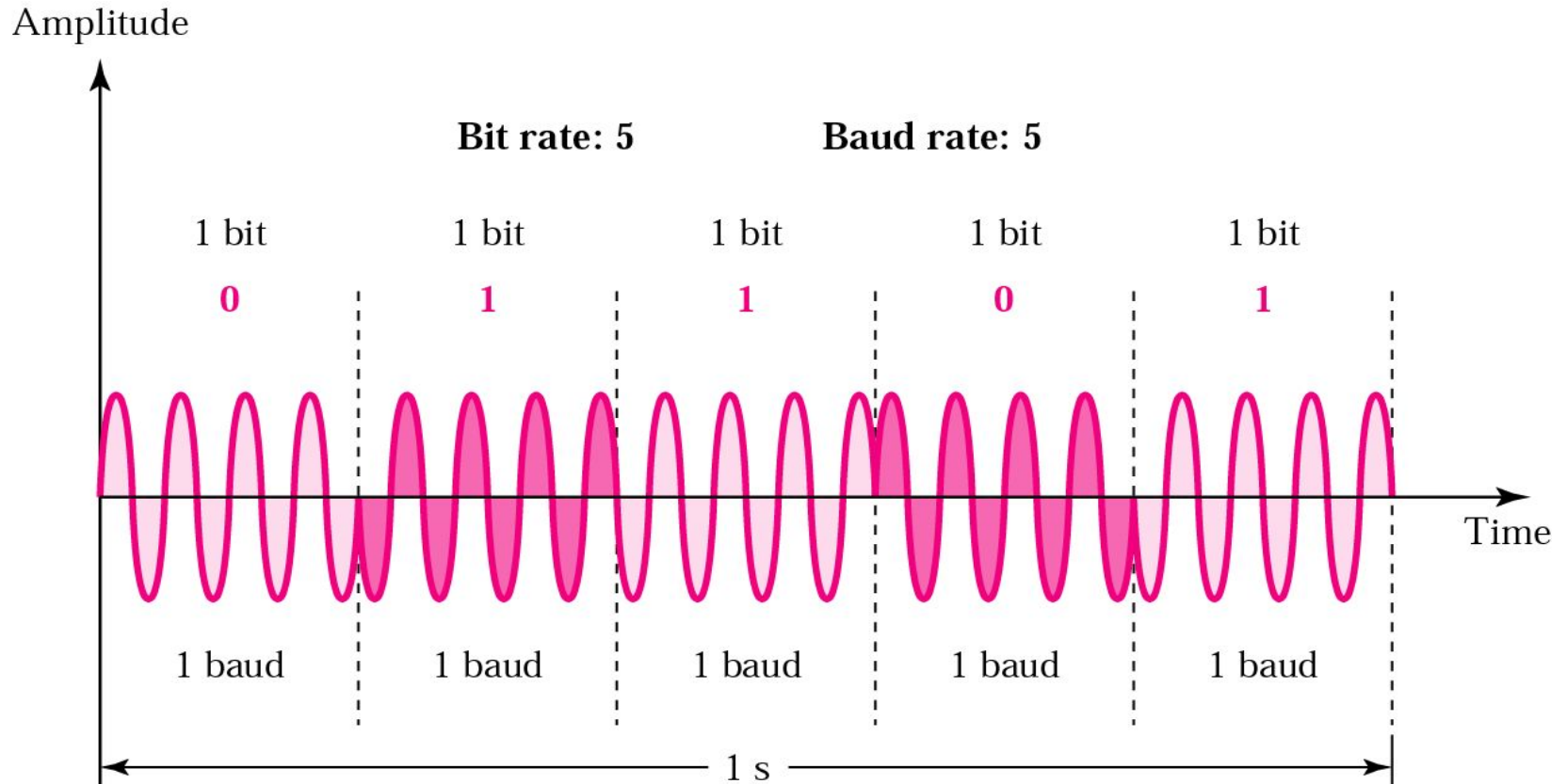
Binary Amplitude Shift Keying



Binary Frequency Shift Keying



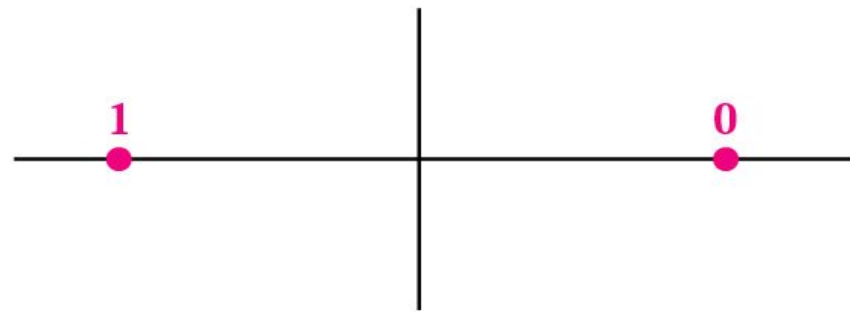
Binary Phase Shift Keying



PSK Constellation

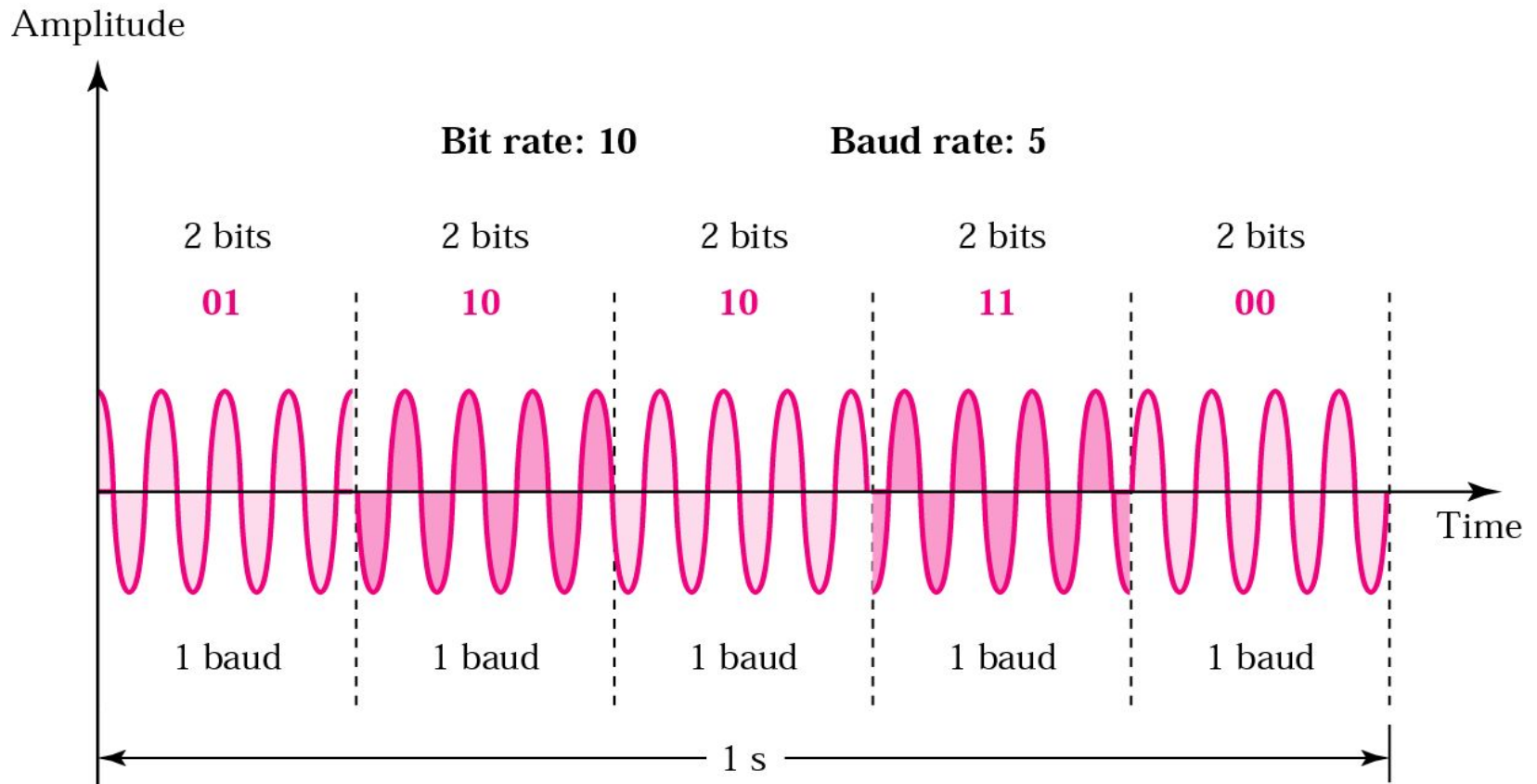
Bit	Phase
0	0
1	180

Bits



Constellation diagram

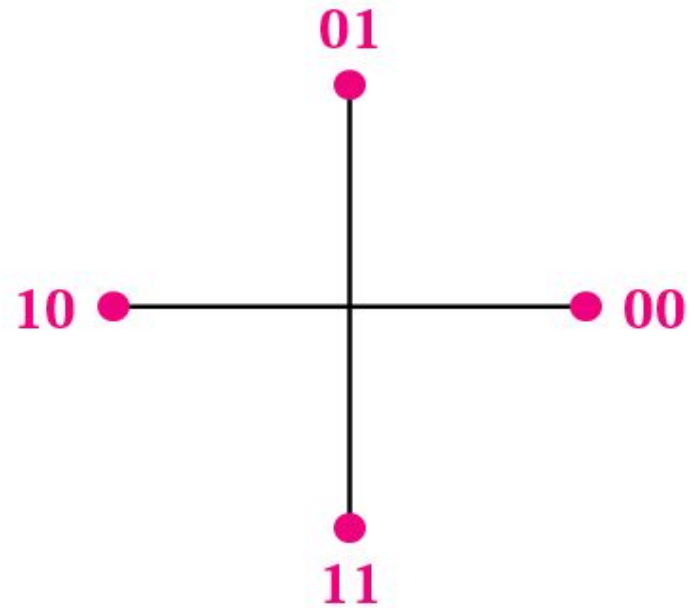
4 PSK Method



4 PSK Constellation

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)

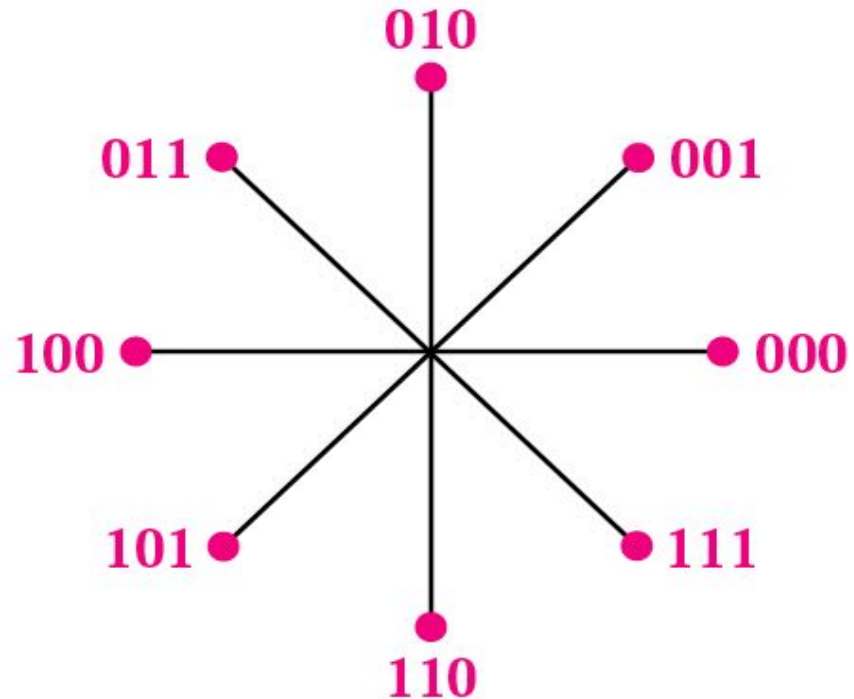


Constellation diagram

8 PSK Constellation

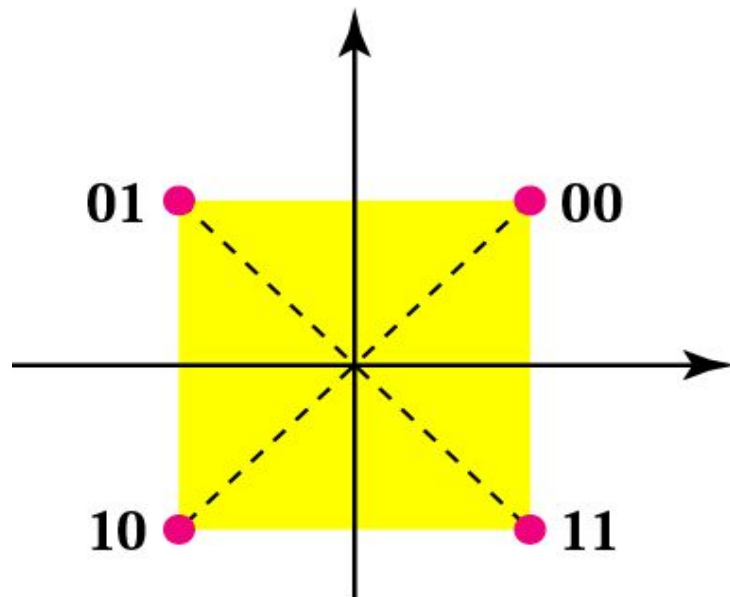
Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

Tribits
(3 bits)

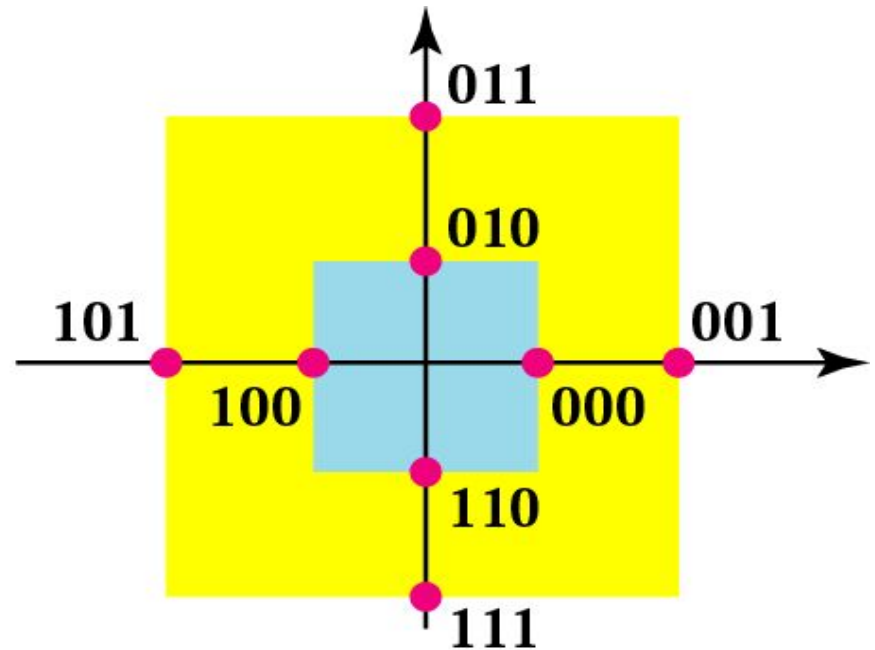


Constellation diagram

4 QAM & 8 QAM

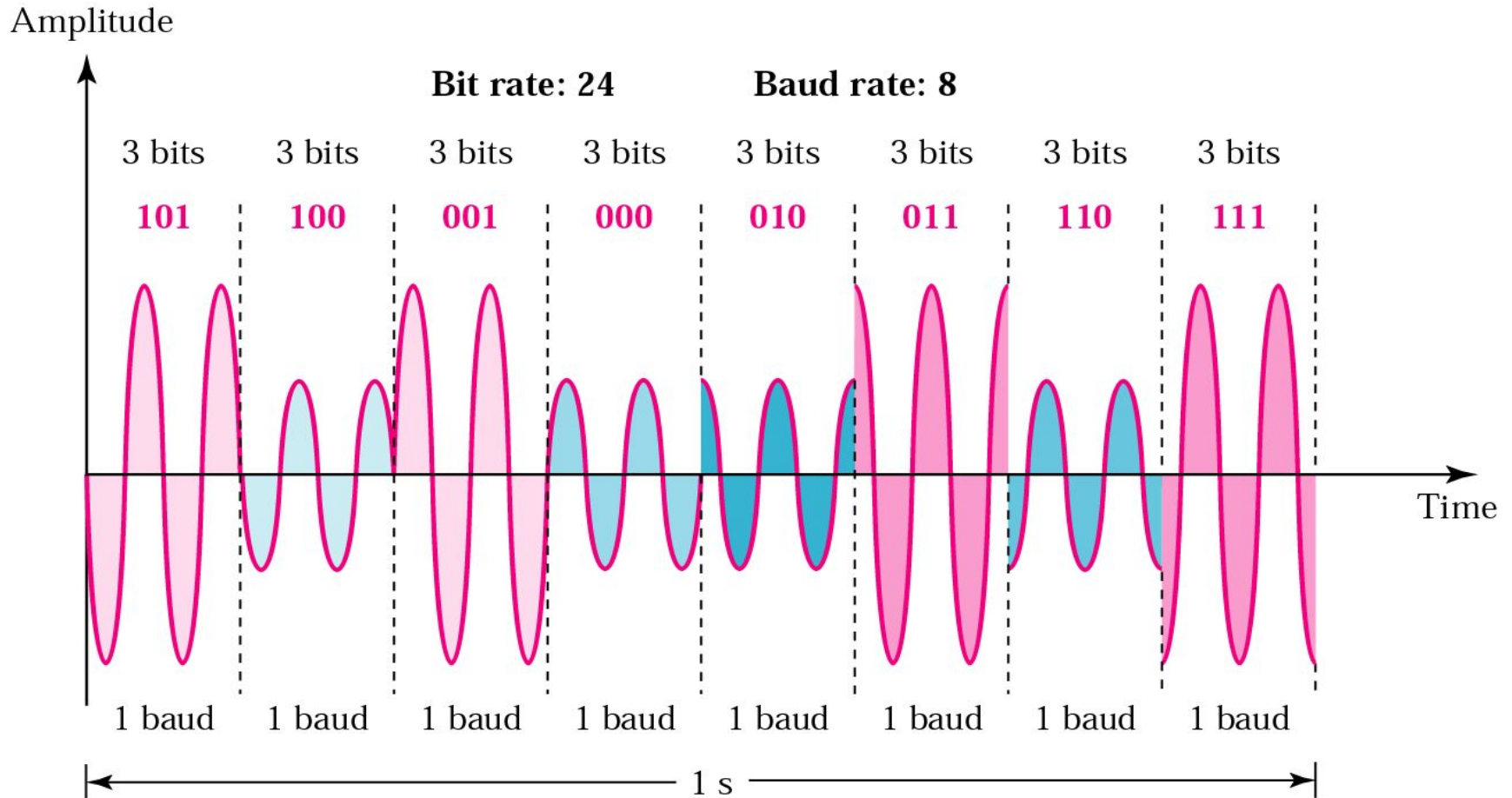


4-QAM
1 amplitude, 4 phases



8-QAM
2 amplitudes, 4 phases

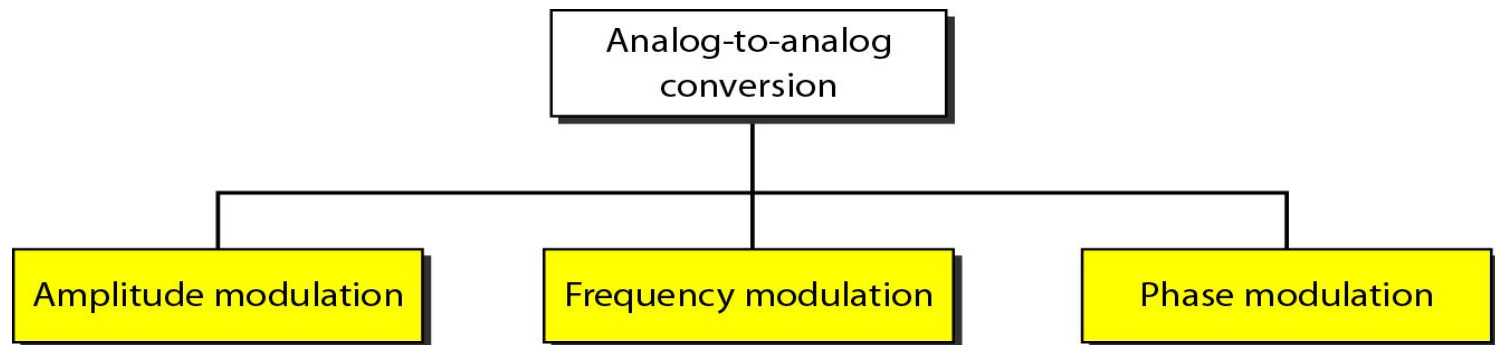
Time Domain for 8 QAM Signal



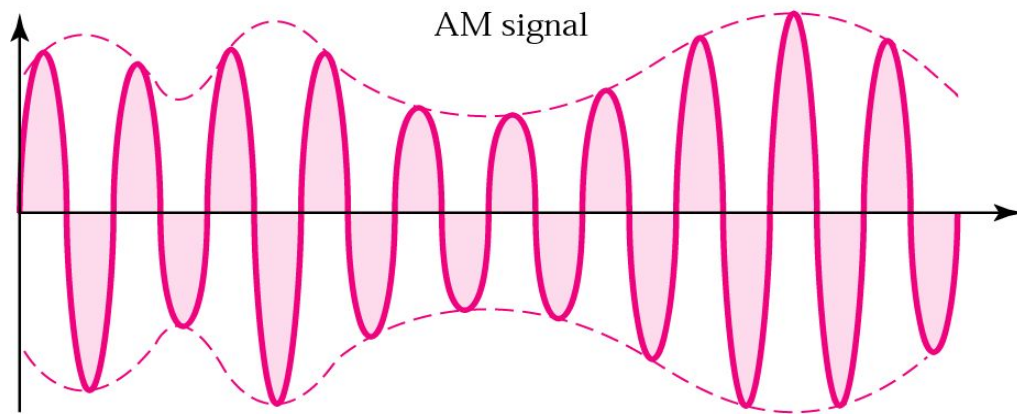
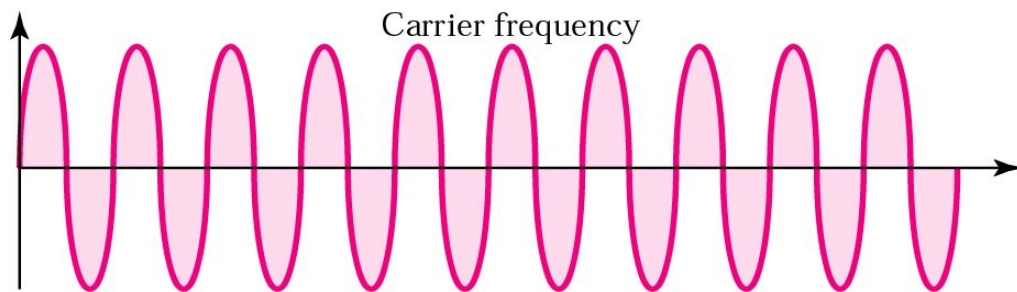
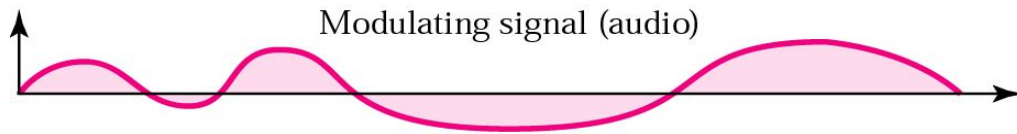
Analog-to-Analog Modulation

Analog-to-Analog

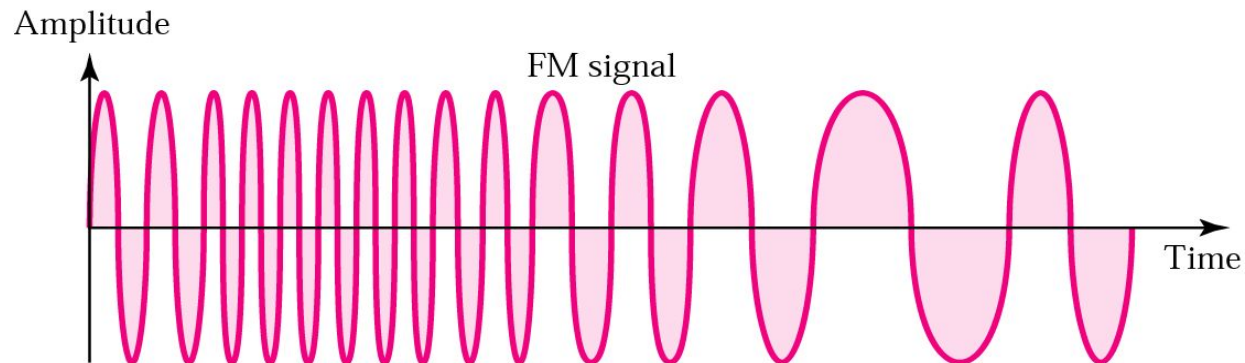
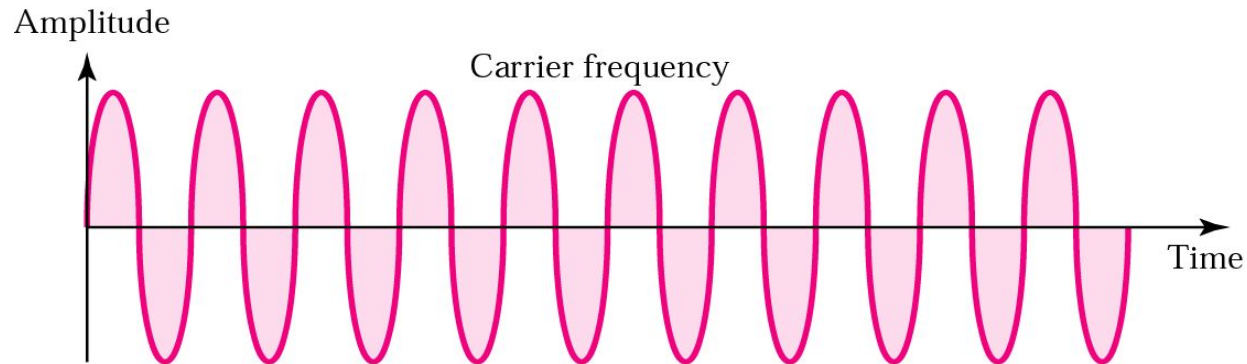
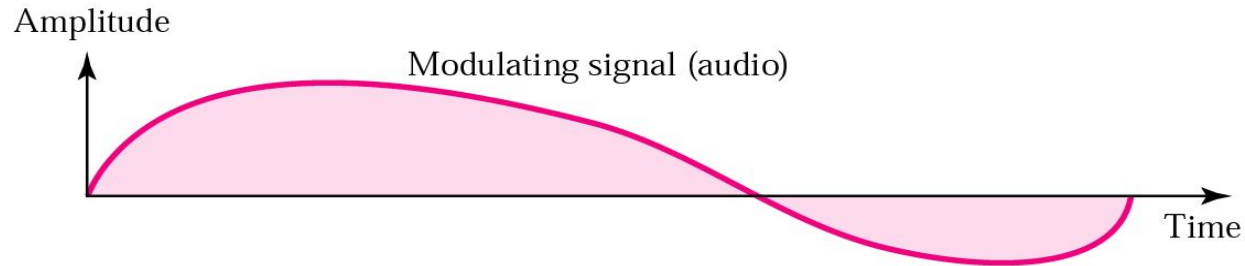
- Analog-to-analog conversion is the representation of analog information by an analog signal. One may ask why we need to modulate an analog signal; it is already analog. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.



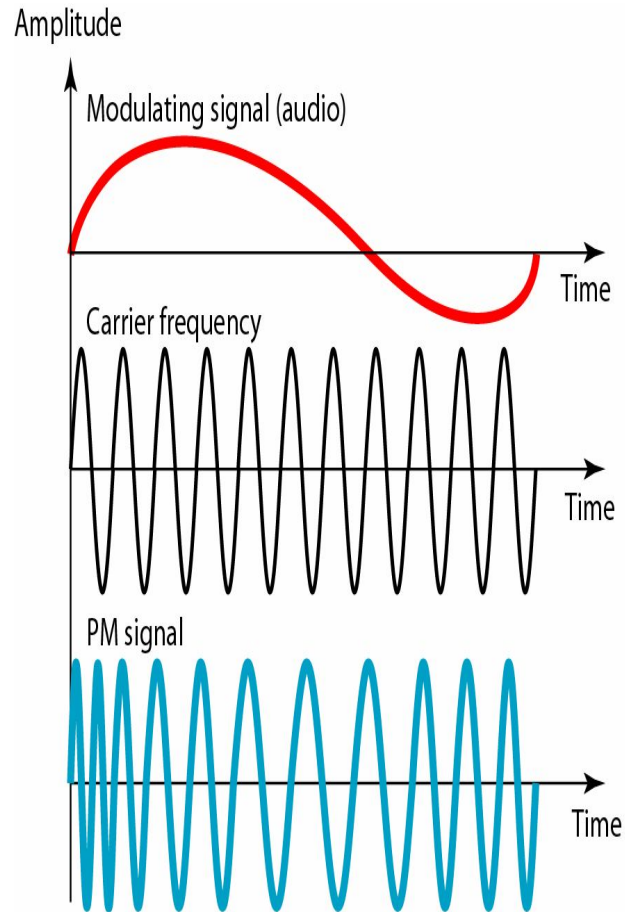
Amplitude Modulation



Frequency Modulation



Phase Modulation



Assignment 2

- You can find Assignment 2 from course web.
- Due Date: First class of Next Week

Quiz 2

- On the day of submission of Assignment related with topics covered in Assignment 2.

Readings

- Chapter 5 (B.A Forouzan)
 - Section 5.1, 5.2

